

Name: _____

Date: _____

Mr. Rawson • Y8 MYP Design



1.3 – You Are The Micro:bit

Unit 1: Robotics & Control

**No laptops today.
Today the class *is* the computer.**

Do Now — from memory with the sheet away

Write what each word means. **Your best four count.**

[4]

input

information going *in* — a button, a shake

output

what comes *out* — the LEDs, a sound

event

something that happens, and starts code running

variable

a named box holding a value you can change

condition

a question with a yes or no answer

algorithm

the steps, in order, that solve the problem

Name: _____

Date: _____

The room is a micro:bit

There are three **containers** on the floor.

Code only runs when it is standing *inside* one.

on start

forever

on button A pressed

on start

runs **once**, the moment the power comes on. Then never again.

forever

runs **over and over**, all lesson, without stopping.

on button A pressed

runs **every time** someone presses A.

The rule that decides everything

A card only runs when **the container it is standing in** is called.

Same cards. Different container. **Completely different program.**

Name: _____

Date: _____

One of you is the variable

The variable is called Steps. One person holds the board and *is* the box.

They do nothing on their own. **They only obey the cards.**

Steps

0

Three cards, three jobs

set Steps to 0

wipe the board and write 0.

change Steps by 1

add one to what is already there.

show number Steps

hold the board up so the room can read it.

Nobody moves until Mr. Rawson calls a container.

Name: _____

Date: _____

Write down what happened

1 The one that works.

on start holds set Steps to 0.

on button A pressed holds two cards, in this order:

change Steps by 1

show number Steps

Button A is pressed four times. Write the number shown each time:

Steps	Steps	Steps	Steps
1	2	3	4

2 Move set Steps to 0 into forever.

Now press A as many times as you like. What does the board show?

always 0 — forever wipes it faster than A can add to it

Why?

forever runs over and over, so set Steps to 0 keeps

happening. The count is wiped before it can ever climb.

Name: _____

Date: _____

3 Move both cards into `on start`.

So `on button A pressed` is now empty.

Power on, then press A four times. What happens?

it shows 1 at power-on, then nothing at all — A does nothing

Why?

`on start` runs once, at power-up. It cannot answer a press.

`on button A pressed` is empty, so there is nothing to run.

Break the counter — teams

Your team gets the **working** program on cards.

1. Move **one card only** into a different container.
2. Write your prediction below. Do not say it out loud.
3. Another team runs it. **1 point** if your prediction was right.

We moved the card: _____

into the container: _____

We predict the board will show:

Name: _____

Date: _____

Before you go

1. A variable is

a box with a name on it, holding a value you can change

2. `on start` and `on button A pressed` are different because

`on start` runs once at power-up; `on button A pressed`

runs every time the button is pressed

3. One thing that is still confusing:

HOMEWORK — The Shake Counter made smaller

Last time this asked for four things at once. **Now it asks for two.**

1. Make a variable called Shakes. Set it to 0 in `on start`.
2. In `on shake`: add 1 to it, then show the number.

That is the whole thing. **No icon. No resetting. No if.** It just counts.

Share → Publish project → submit the URL to **Google Classroom.**

Due Friday 25 September.

Only if you get properly stuck: micro:bit have written this exact project up, step by step, at

microbit.org/projects/make-it-code-it/step-counter

Try it yourself first. Reading the answer is not the same as knowing it.